# Run report: EXP-SEMBIN-24ec56

This report records what ran and what it measured. It is generated by tools/runbook.py from the session journal. It states no verdict, support, weakening or significance; the evidence review does that.

- Runbook: experiments/EXP-SEMBIN-24ec56/runbook.yaml (sha256 2996f44f854ecb70712ddca4ef64e31888467cdaa37dbea739824c717d38fd5c)
- Session: experiments/EXP-SEMBIN-24ec56/runbook-sessions/RBS-SEMBIN-24ec56
- Journal tip: eb3ee32b5ef7c997fa96fc7630e4beace23238f6004fe0452cfd447e8b959030 (40 entries)
- Commit: 5ac50196675bf0cdd7195c7a74863bed8473e231; tracked files dirty: False
- Started 2026-10-26T21:02:28.326405Z, finished 2026-10-26T21:02:42.564962Z; status stopped

## Purpose

Measures, per cell (m in 3..5; n, k per the plan; V polynomial and random), the share of enumeration-certified UNSAT chained-S_3 descended systems refuted by the degree-4 mutant closure W_4 (with its Clopper-Pearson 95% interval), the false-refutation count on SAT instances, the plain Macaulay M_4 refutation count and the D_W distribution, beside the same statistics on support-matched and dense nulls, the m=2 control, the block-local L2adj1 arm at m=4, the C1 pilot gate and the C6 determinism replicate, with two GF(2) eliminators and two enumerators. Observations only; check.py re-derives every count from instances[] and re-labels a seeded sample from source. No speedup, no cost against rho and no statement at n = 131 is claimed. One-cell successor of EXP-SEMBIN-d9767f: the parent's interrupted RUN-SEMBIN-07f442 is preserved there as a non-evidence attempt.

## Steps

| step | status | what it does |
|---|---|---|
| S01-trials | failed | Runs run.py then check.py for every trial of trial-plan.json under the supervisor claim and keeps every output file. |

## Units

0 valid of 36; invalid: ['ctrl-c1-gate-m3-n9', 'ctrl-c6-replicate-m3-n9', 'ctrl-m2-n11', 'ctrl-m2-n9', 'null-dense-m3-n11', 'null-dense-m3-n13', 'null-dense-m3-n15', 'null-dense-m3-n9', 'null-dense-m4-n9', 'null-support-m3-n11', 'null-support-m3-n13', 'null-support-m3-n15', 'null-support-m3-n9', 'null-support-m4-n11', 'null-support-m4-n13', 'null-support-m4-n9', 'null-support-m5-n11', 'null-support-m5-n9', 'poly-m3-n11', 'poly-m3-n13', 'poly-m3-n15', 'poly-m3-n9', 'poly-m4-n11', 'poly-m4-n13', 'poly-m4-n9', 'poly-m5-n11', 'poly-m5-n9', 'rand-m3-n11', 'rand-m3-n13', 'rand-m3-n15', 'rand-m3-n9', 'rand-m4-n11', 'rand-m4-n13', 'rand-m4-n9', 'rand-m5-n11', 'rand-m5-n9']

| trial | run | receipt status | valid |
|---|---|---|---|
| ctrl-c1-gate-m3-n9 (control) | RUN-SEMBIN-024883 | not_run | False |
| poly-m3-n9 | RUN-SEMBIN-0689a2 | not_run | False |
| ctrl-c6-replicate-m3-n9 (control) | RUN-SEMBIN-1458e0 | not_run | False |
| ctrl-m2-n9 (control) | RUN-SEMBIN-25ef78 | not_run | False |
| ctrl-m2-n11 (control) | RUN-SEMBIN-2701bd | not_run | False |
| rand-m3-n9 | RUN-SEMBIN-2f75c2 | not_run | False |
| poly-m3-n11 | RUN-SEMBIN-31542d | not_run | False |
| rand-m3-n11 | RUN-SEMBIN-35a6ff | not_run | False |
| null-support-m3-n9 (control) | RUN-SEMBIN-41a48a | not_run | False |
| null-support-m3-n11 (control) | RUN-SEMBIN-521fd9 | not_run | False |
| null-dense-m3-n9 (control) | RUN-SEMBIN-5ad0a7 | not_run | False |
| null-dense-m3-n11 (control) | RUN-SEMBIN-636347 | not_run | False |
| poly-m4-n9 | RUN-SEMBIN-63c142 | not_run | False |
| rand-m4-n9 | RUN-SEMBIN-6718c2 | not_run | False |
| null-support-m4-n9 (control) | RUN-SEMBIN-6928a2 | not_run | False |
| null-dense-m4-n9 (control) | RUN-SEMBIN-79be6e | not_run | False |

| | | | |
|---|---|---|---|
| poly-m3-n15 | RUN-SEMBIN-7b3518 | not_run | False |
| rand-m3-n15 | RUN-SEMBIN-87be3b | not_run | False |
| null-support-m3-n15 (control) | RUN-SEMBIN-886cb7 | not_run | False |
| null-dense-m3-n15 (control) | RUN-SEMBIN-8ae959 | not_run | False |
| poly-m3-n13 | RUN-SEMBIN-8bd3c8 | not_run | False |
| rand-m3-n13 | RUN-SEMBIN-989e60 | not_run | False |
| null-support-m3-n13 (control) | RUN-SEMBIN-9f8171 | not_run | False |
| null-dense-m3-n13 (control) | RUN-SEMBIN-9fed7b | not_run | False |
| poly-m4-n11 | RUN-SEMBIN-a469b6 | not_run | False |
| rand-m4-n11 | RUN-SEMBIN-bd0737 | not_run | False |
| null-support-m4-n11 (control) | RUN-SEMBIN-bf1eb8 | not_run | False |
| poly-m5-n9 | RUN-SEMBIN-c3dc77 | not_run | False |
| rand-m5-n9 | RUN-SEMBIN-c45847 | not_run | False |
| null-support-m5-n9 (control) | RUN-SEMBIN-d160cd | not_run | False |
| poly-m4-n13 | RUN-SEMBIN-d3b5e1 | not_run | False |
| rand-m4-n13 | RUN-SEMBIN-e8d008 | not_run | False |
| null-support-m4-n13 (control) | RUN-SEMBIN-f03b11 | not_run | False |
| poly-m5-n11 | RUN-SEMBIN-f648a3 | not_run | False |
| rand-m5-n11 | RUN-SEMBIN-f883b4 | not_run | False |
| null-support-m5-n11 (control) | RUN-SEMBIN-fa6a48 | not_run | False |

## Metrics

Each value is computed by tools/runbook.py from the recorded outputs named in its source. Labels say whether a value was measured or derived from measured values.

| metric | label | unit | n | values |
|---|---|---|---|---|
| censored_jobs | measured | jobs | 0 | count=0, max=null, sum=null |
| elapsed_seconds | measured | seconds | 0 | count=0, max=null, mean=null |
| eliminator_disagreements | measured | pairs | 0 | count=0, max=null, sum=null |
| false_refutations | measured | instances | 0 | count=0, max=null, sum=null |
| m4_refutations | measured | instances | 0 | count=0, max=null, sum=null |
| r4_m3_n11_k4_poly | measured | fraction | 0 | count=0, max=null, min=null |
| r4_m3_n11_k4_rand | measured | fraction | 0 | count=0, max=null, min=null |
| r4_m3_n13_k5_poly | measured | fraction | 0 | count=0, max=null, min=null |
| r4_m3_n13_k5_rand | measured | fraction | 0 | count=0, max=null, min=null |
| r4_m3_n15_k5_poly | measured | fraction | 0 | count=0, max=null, min=null |
| r4_m3_n15_k5_rand | measured | fraction | 0 | count=0, max=null, min=null |
| r4_m3_n9_k3_poly | measured | fraction | 0 | count=0, max=null, min=null |
| r4_m3_n9_k3_rand | measured | fraction | 0 | count=0, max=null, min=null |
| r4_m4_n11_k3_poly | measured | fraction | 0 | count=0, max=null, min=null |
| r4_m4_n11_k3_rand | measured | fraction | 0 | count=0, max=null, min=null |
| r4_m4_n13_k4_poly | measured | fraction | 0 | count=0, max=null, min=null |
| r4_m4_n13_k4_rand | measured | fraction | 0 | count=0, max=null, min=null |
| r4_m4_n9_k3_poly | measured | fraction | 0 | count=0, max=null, min=null |
| r4_m4_n9_k3_rand | measured | fraction | 0 | count=0, max=null, min=null |

| | | | | |
|---|---|---|---|---|
| r4_m5_n11_k3_poly | measured | fraction | 0 | count=0, max=null, min=null |
| r4_m5_n11_k3_rand | measured | fraction | 0 | count=0, max=null, min=null |
| r4_m5_n9_k3_poly | measured | fraction | 0 | count=0, max=null, min=null |
| r4_m5_n9_k3_rand | measured | fraction | 0 | count=0, max=null, min=null |
| unsat_w4_run_m3_n11_k4_poly | measured | instances | 0 | count=0, max=null, min=null |
| unsat_w4_run_m3_n11_k4_rand | measured | instances | 0 | count=0, max=null, min=null |
| unsat_w4_run_m3_n13_k5_poly | measured | instances | 0 | count=0, max=null, min=null |
| unsat_w4_run_m3_n13_k5_rand | measured | instances | 0 | count=0, max=null, min=null |
| unsat_w4_run_m3_n15_k5_poly | measured | instances | 0 | count=0, max=null, min=null |
| unsat_w4_run_m3_n15_k5_rand | measured | instances | 0 | count=0, max=null, min=null |
| unsat_w4_run_m3_n9_k3_poly | measured | instances | 0 | count=0, max=null, min=null |
| unsat_w4_run_m3_n9_k3_rand | measured | instances | 0 | count=0, max=null, min=null |
| unsat_w4_run_m4_n11_k3_poly | measured | instances | 0 | count=0, max=null, min=null |
| unsat_w4_run_m4_n11_k3_rand | measured | instances | 0 | count=0, max=null, min=null |
| unsat_w4_run_m4_n13_k4_poly | measured | instances | 0 | count=0, max=null, min=null |
| unsat_w4_run_m4_n13_k4_rand | measured | instances | 0 | count=0, max=null, min=null |
| unsat_w4_run_m4_n9_k3_poly | measured | instances | 0 | count=0, max=null, min=null |
| unsat_w4_run_m4_n9_k3_rand | measured | instances | 0 | count=0, max=null, min=null |
| unsat_w4_run_m5_n11_k3_poly | measured | instances | 0 | count=0, max=null, min=null |
| unsat_w4_run_m5_n11_k3_rand | measured | instances | 0 | count=0, max=null, min=null |
| unsat_w4_run_m5_n9_k3_poly | measured | instances | 0 | count=0, max=null, min=null |
| unsat_w4_run_m5_n9_k3_rand | measured | instances | 0 | count=0, max=null, min=null |

EXP-SEMBIN-24ec56: steps as recorded (fewer than two comparable values to plot)

S01-trials: failed

Figure 1: Recorded values from this session only; no new computation.

## Validity, proofs and formal obligations

- Execution validity: incomplete
- Control validity: failed
- Inferential validity: not_assessed: review-evidence and the Coordinator decide
- Execution proofs: {"chain": "incomplete: docs/execution-proof-chain.md milestones are open", "failed_units": 0, "unavailable_units": 0, "verified_units": 0}
- Audit: not audited: tools/runbook.py audit (audit-runs skill)

## Where it stopped

S01-trials: infrastructure_error: supervisor exit 2 (impediment: the supervisor refused to launch); see experiments/EXP-SEMBIN-24ec56/runbook-sessions/RBS-SEMBIN-24ec56/steps/S01-trials.attempt-1.supervisor.stderr

A stop is an operational fact. It is not evidence about the hypothesis.